

- Description

This module manages a smart cold storage temperature monitoring and cooling control system, ensuring that:

1. **The temperature inside the cold storage is continuously monitored to prevent spoilage of stored goods.**
2. **If the temperature rises above the maximum safe threshold, the cooling system is activated.**
3. **If the temperature drops below the minimum threshold, the cooling system is deactivated to save energy.**
4. **A log of all temperature readings, cooling system activations, and shutdowns is maintained for operational efficiency.**

This system helps maintain product freshness, reduce energy consumption, and automate climate control in refrigerated storage facilities.

- Specification

module SmartColdStorageControl

var currentTemperature, maxTemperature, minTemperature: real

var coolingSystemActive: boolean

process updateTemperature(changeAmount: real)

functional scenario form:

currentTemperature <= maxTemperature && Storage temperature maintained.

||

currentTemperature > maxTemperature && coolingSystemActive = true,

Cooling system activated.

||

currentTemperature < minTemperature && coolingSystemActive = false,

Cooling system deactivated.

end process

end module

- Java Code

```
package code;
class SmartColdStorage {
    private double maxTemperature;
    private double minTemperature;
    private double currentTemperature;
    private boolean coolingSystemActive;
    public SmartColdStorage(double maxTemperature, double minTemperature, double initialTemperature) {
        this.maxTemperature = maxTemperature;
        this.minTemperature = minTemperature;
        this.currentTemperature = initialTemperature;
        this.coolingSystemActive = false;
    }
}
```

```

    }
    public void updateTemperature(double changeAmount) {
        currentTemperature += changeAmount;
        manageCoolingSystem();
        System.out.printf("Current Storage Temperature: %.2f°C\n", currentTemperature);
        if (coolingSystemActive) {
            System.out.println("Cooling System Activated.");
        } else {
            System.out.println("Cooling System Deactivated.");
        }
    }
    private void manageCoolingSystem() {
        if (currentTemperature > maxTemperature) {
            coolingSystemActive = true;
        } else if (currentTemperature < minTemperature) {
            coolingSystemActive = false;
        }
    }
}

public class c65_SmartColdStorageSystem {
    public static void main(String[] args) {
        SmartColdStorage coldStorage = new SmartColdStorage(5.0, -5.0, 0.0);

        // testcase-VT:
        // coldStorage.updateTemperature(6.0);
        // coldStorage.updateTemperature(-4.0);
        // coldStorage.updateTemperature(-7.0);
        // coldStorage.updateTemperature(-12.0);
        // coldStorage.updateTemperature(3.0);

        // testcase-FT:
        // coldStorage.updateTemperature(1.6349721113176986E296);
        // coldStorage.updateTemperature(-1.7976931348623157E308);
        // coldStorage.updateTemperature(1.0222176649217088E308);
        // coldStorage.updateTemperature(1.0432843813458766E308);
        // coldStorage.updateTemperature(0.0);

        // testcase-Z3:
        // coldStorage.updateTemperature(6.0);
        // coldStorage.updateTemperature(-4.0);
        // coldStorage.updateTemperature(-7.0);
        // coldStorage.updateTemperature(-12.0);
        // coldStorage.updateTemperature(3.0);
    }
}

```

```

// testcase-UVT:
coldStorage.updateTemperature(6.0);
coldStorage.updateTemperature(-4.0);
coldStorage.updateTemperature(-7.0);
coldStorage.updateTemperature(-12.0);
coldStorage.updateTemperature(3.0);
}
}

```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c65_SmartColdStorageSystem.java	100.0 %	6	0	6
> c65_SmartColdStorageSystem		0	0	0
> SmartColdStorage	100.0 %	6	0	6

3. Vibration testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	20,702	20,791
▼ src	0.4 %	89	20,702	20,791
▼ code	0.4 %	89	20,702	20,791
▼ c65_SmartColdStorageSystem.java	96.7 %	89	3	92
> c65_SmartColdStorageSystem	88.5 %	23	3	26
> SmartColdStorage	100.0 %	66	0	66

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c65_SmartColdStorageSystem.java	83.3 %	5	1	6
> c65_SmartColdStorageSystem		0	0	0
> SmartColdStorage	83.3 %	5	1	6

5. Fuzz testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	20,702	20,791
▼ src	0.4 %	89	20,702	20,791
▼ code	0.4 %	89	20,702	20,791
▼ c65_SmartColdStorageSystem.java	96.7 %	89	3	92
> c65_SmartColdStorageSystem	88.5 %	23	3	26
> SmartColdStorage	100.0 %	66	0	66

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c65_SmartColdStorageSystem.java	100.0 %	6	0	6
> c65_SmartColdStorageSystem		0	0	0
> SmartColdStorage	100.0 %	6	0	6

7. Z3 instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
^				
▼ VibrationTestingTool	0.4 %	89	20,702	20,791
▼ src	0.4 %	89	20,702	20,791
▼ code	0.4 %	89	20,702	20,791
▼ c65_SmartColdStorageSystem.java	96.7 %	89	3	92
> c65_SmartColdStorageSystem	88.5 %	23	3	26
> SmartColdStorage	100.0 %	66	0	66

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
^				
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c65_SmartColdStorageSystem.java	100.0 %	6	0	6
> c65_SmartColdStorageSystem		0	0	0
> SmartColdStorage	100.0 %	6	0	6

9. Unimproved V-method instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
^				
▼ VibrationTestingTool	0.4 %	89	20,702	20,791
▼ src	0.4 %	89	20,702	20,791
▼ code	0.4 %	89	20,702	20,791
▼ c65_SmartColdStorageSystem.java	96.7 %	89	3	92
> c65_SmartColdStorageSystem	88.5 %	23	3	26
> SmartColdStorage	100.0 %	66	0	66